Multi-Image AI Question Answering System

1. Introduction

With the advancement of Artificial Intelligence, systems are now capable of understanding both images and text simultaneously. This project presents a **Multi-Image AI Question Answering System** that allows users to upload multiple images and ask questions related to them.

The system uses a **multimodal AI model (Gemini)** to interpret visual data and generate intelligent responses. It also includes mechanisms to ensure robustness, safety, and performance monitoring.

2. Problem Statement

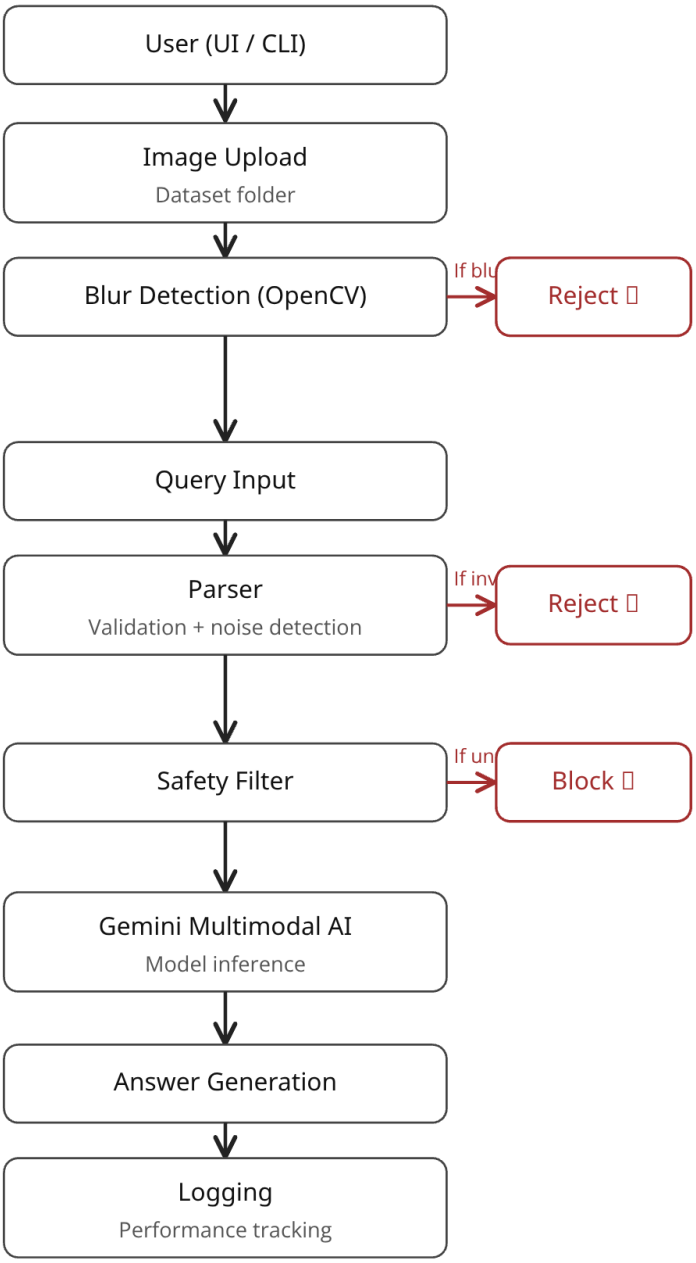
Many AI systems fail in real-world scenarios due to:

- Noisy inputs (e.g., $$$$, 1e12)
- Poor-quality images (blurred images)
- Irrelevant or unsafe queries
- Lack of monitoring and logging

This project aims to build a system that:

- Answers questions strictly based on images
- Handles invalid inputs effectively
- Prevents unsafe or sensitive queries
- Maintains system stability and logs performance

3. System Architecture



4. Workflow Sequence

1. User uploads images or system loads images from folder
2. Each image is checked for blur (low-quality images are removed)
3. User enters a query
4. Query is validated using parser
5. Safety check blocks restricted queries

6. Valid input is sent to Gemini AI model
7. AI generates answer based on images
8. Response is displayed to user
9. Query, response, and performance are logged

5. Technologies Used

- **Python** – Core programming language
- **OpenCV (cv2)** – Blur detection
- **Google Generative AI (Gemini)** – Multimodal AI model
- **dotenv** – API key management
- **React (Frontend)** – User interface

6. Key Components

6.1 Query Parser

- Filters invalid inputs
- Detects noise (symbols, scientific notation)
- Allows only image-related queries

6.2 Blur Detection

- Uses Laplacian variance method
- Removes low-quality images
- Improves accuracy

6.3 Safety Filter

Blocks sensitive queries like:

- "Who is this person"
- "Identify this person"

6.4 Gemini AI Model

- Processes both images and text
- Generates intelligent responses
- Multimodal reasoning

6.5 Logging System

Stores:

- Queries
- Answers
- Errors
- Performance

7. Testing & Evaluation (Day 13)

Test Case	Input	Result
Normal Query	What is in the image	Correct
Noise Input	$$$$, 1e12	Rejected
Non-image Query	What is Python	Rejected
Safety Test	Identify person	Blocked
Blur Image	Blurry image	Skipped
Stress Test	Multiple queries	Stable

8. Performance Results (Day 14)

- Average Response Time: **1–3 seconds**
- System Stability: **No crashes observed**
- Accuracy: **High for general queries**
- Robustness: **Handles noise and edge cases effectively**

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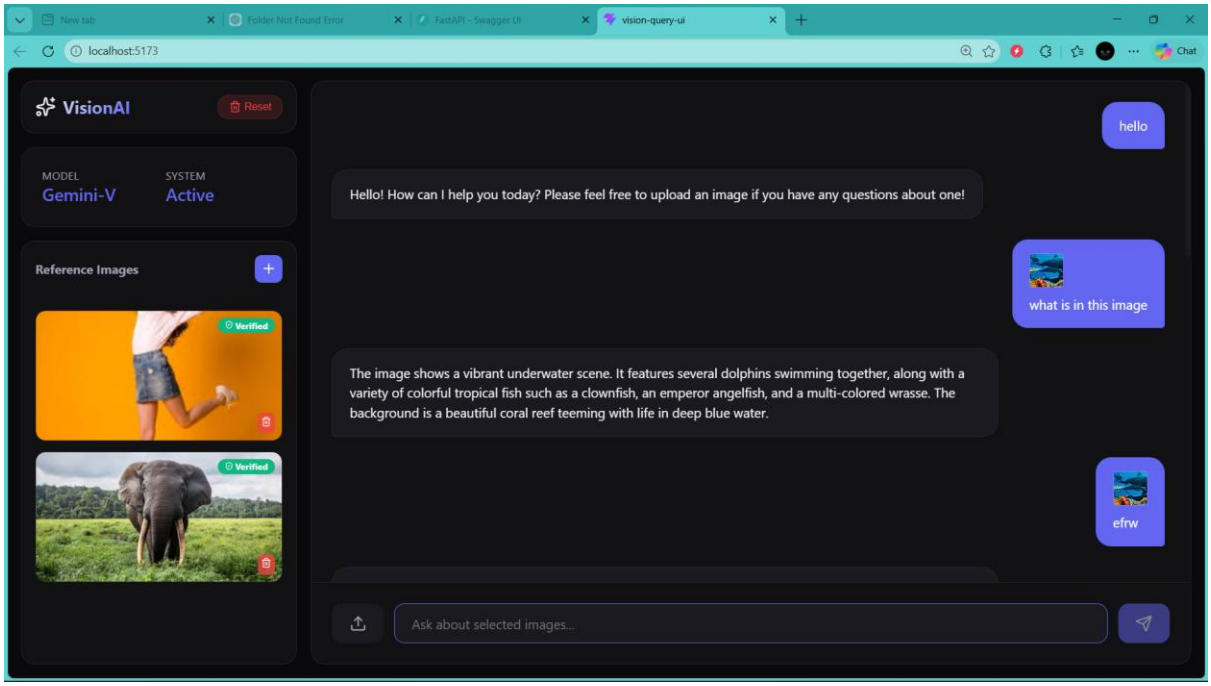
9. Project Structure

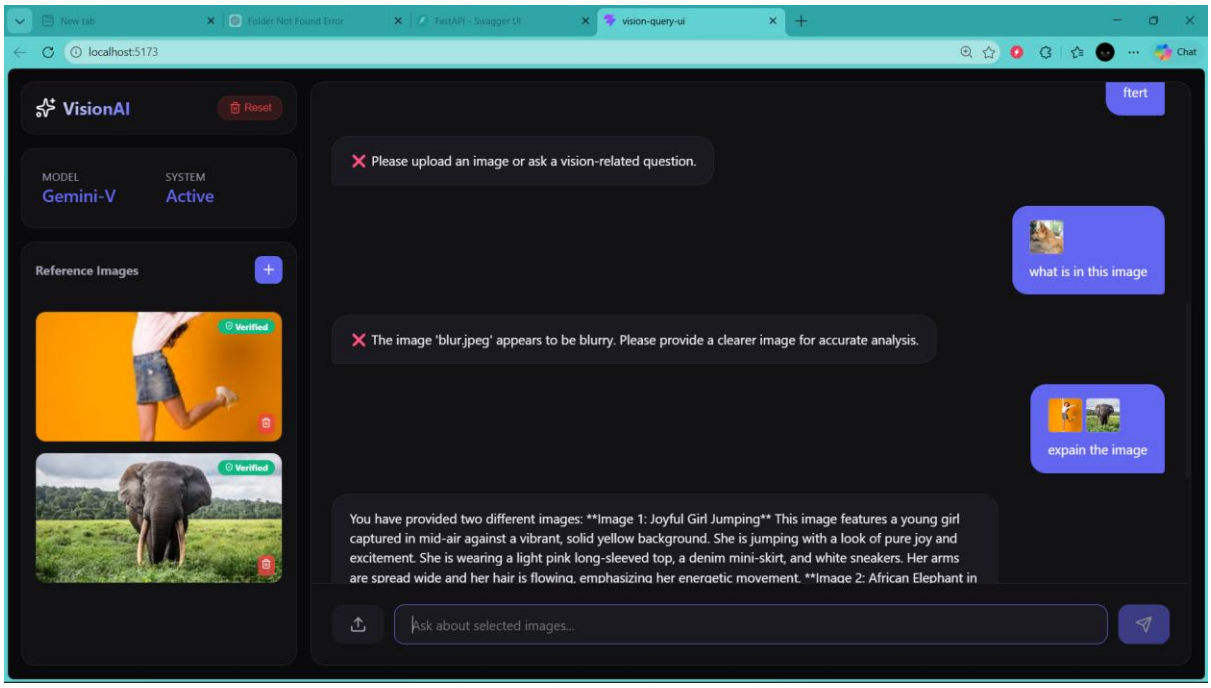
```
capston_project/
|
|__backend
    |__main.py            # Agent entry point (full pipeline)
 ├── .env                # API key (NOT committed to git)
 ├── images/             # Folder containing input images (.jpg / .png)
|   └── sample.jpg
|__vision-query-ui
    |__src
      |__ App.jsx   # ui code
 ├── requirements.txt      # All dependencies
 ├── README.md             # This file
```

10. Features

- Multi-image support
- AI-based image understanding
- Noise and invalid input detection
- Blur detection
- Safety filtering
- Logging system
- Performance tracking
- React-based UI

11.Output


New tab
Folder Not Found Error
FastAPI - Swagger UI
vision-query-ui
localhost:5173
Chat
VisionAI
Reset
MODEL
Gemini-V
SYSTEM
Active
Reference Images
Verified
Verified
hello
Hello! How can I help you today? Please feel free to upload an image if you have any questions about one!
what is in this image
The image shows a vibrant underwater scene. It features several dolphins swimming together, along with a variety of colorful tropical fish such as a clownfish, an emperor angelfish, and a multi-colored wrasse. The background is a beautiful coral reef teeming with life in deep blue water.
efrw
Ask about selected images...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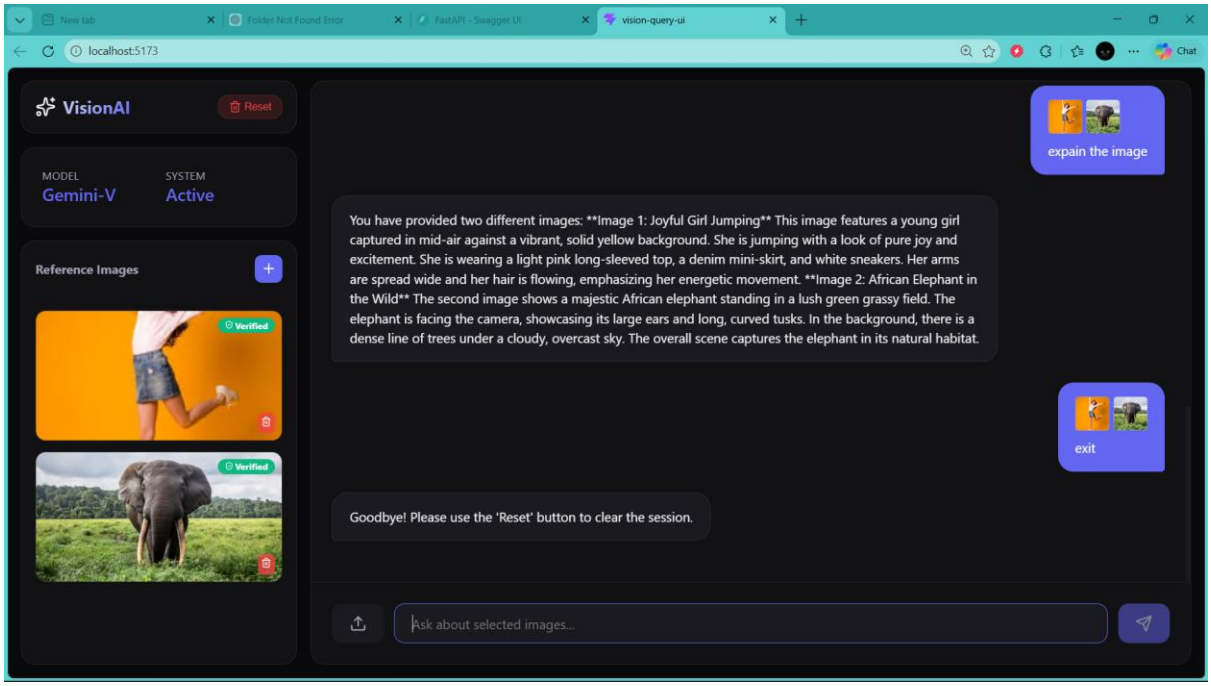
Reference Images
Verified
Verified
ftert
Please upload an image or ask a vision-related question.
what is in this image
The image 'blur.jpeg' appears to be blurry. Please provide a clearer image for accurate analysis.
expain the image
You have provided two different images: **Image 1: Joyful Girl Jumping** This image features a young girl captured in mid-air against a vibrant, solid yellow background. She is jumping with a look of pure joy and excitement. She is wearing a light pink long-sleeved top, a denim mini-skirt, and white sneakers. Her arms are spread wide and her hair is flowing, emphasizing her energetic movement. **Image 2: African Elephant in
Ask about selected images...



12. Conclusion

This project successfully demonstrates a robust AI system capable of understanding images and answering related queries. The system is designed with strong validation, safety, and monitoring features, making it reliable and production-ready.